

DIY Drone Build — Indoor 3DGS Scanning

Goal: minimal assembly, max programming freedom, indoor flat/apartment capture.

Core build

Item	Model	Price	Source
Frame + flight controller	Holybro X500 V2 PX4 Dev Kit	\$579 (~2,900–3,100 zł landed)	Holybro direct — ships outside EU, PL VAT + duty applies
Companion computer	NVIDIA Jetson Orin Nano Super (8GB)	1,600 zł	Botland (Kraków) — confirmed local stock, no import
LiDAR	Livox Mid-360 (360°, 1–2cm accuracy)	~3,000–3,300 zł	owlrobotics (Germany, EU) — use business VAT ID for reverse charge
Camera	e-con Systems See3CAM_130 (13MP, USB3)	~1,200 zł (est., not yet sourced)	TBD
Battery	4S 5000mAh LiPo	~200–220 zł	Local RC shop
Charger	4S balance charger	~220–250 zł	Local RC shop
Mounting/cabling	standoffs, Mid-360 cable, connectors	~200–250 zł	Mixed
Prop guards	indoor-safe set for X500	~150–200 zł	Local RC shop / AliExpress

Total: ~ 9,200–10,300 zł (~\$2,300–2,600 / €2,150–2,450)

Software stack

- Flight: PX4 (stock, on the Pixhawk 6C — don't modify)
- Companion: Jetson runs Livox SDK2 + FAST-LIO2 (SLAM/obstacle point cloud) +

camera capture, hardware-triggered/timestamp-synced to LiDAR

- Output: raw frame sequences + pose + LiDAR point cloud → COLMAP / Brush

Cheaper alternative (if starting small)

Swap Mid-360 → **Intel RealSense D435i** (~1,300 zł, officially PX4-Avoidance supported, forward-facing only, real blind-spot risk indoors). Saves ~1,900 zł. New total ≈ 7,900 zł.

Rejected/parked options

- **Antigravity A1** (\$1,279–1,599, ready-to-fly): wrong tool — stitched/inpainted 360 footage breaks SfM reconstruction, closed ecosystem, no raw data access. Possibly useful separately for marketing hero shots only.
- **Phone (ARKit) as sole sensor/FC input**: technically possible (PX4 accepts any MAVLink `ODOMETRY` source) but no ready bridge exists for iPhone, single-direction blind spot, iOS blocks easy wired MAVLink. Parked — not worth the engineering risk yet.
- **Handheld LiDAR scanner** (Mandeye/OpenSLAM-style, same Mid-360 + tiny SBC + 3D-printed grip, ~€2,050): may replace/complement the drone specifically for **interiors** — walking a handheld rig through rooms avoids crash risk and reaches tight spots a drone can't. Worth building as a second rig using the same Mid-360, keep drone for exteriors.
- **2D spinning LiDAR / toy-drone IR sensors**: too sparse (single plane or single point), not viable for 3D indoor mapping.

Still unresolved

- Exact camera SKU/price not confirmed
- Whether to buy Mid-360 as business (VAT reverse charge) or personal